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Automated Workflow for Optimizing CO₂ Storage Well Placement by Coupling Static Properties and Dynamic Simulation

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Abstract

Carbon Capture and Storage (CCS) is a vital technology for reducing greenhouse gas emissions and mitigating climate change. Saline aquifers have emerged as a promising solution for large-scale CO₂ sequestration, offering significant storage potential. However, the complex interplay of geological factors, including reservoir heterogeneity, spill point depth, and reservoir pressure, poses a significant challenge in optimizing well placement. To overcome this challenge, we have developed a cutting-edge automation workflow that combines static model properties and dynamic simulations to determine optimal well locations and completion strategies. By harnessing the power of this integrated simulation approach, our workflow seeks to maximize storage efficiency, minimize well injectivity issues and enhance overall CCS effectiveness.

This methodology automates the optimization of CO₂ injector placement and completion by integrating reservoir static and dynamic variables to identify the most suitable areas. The workflow involves a series of steps, including screening for high-permeability zones, connected volumes, favorable reservoir pressure, as well as spill point depth, to generate optimal "sweet spots". Wells are then strategically placed and completed at these identified sweet spots, followed by dynamic simulations to compute storage efficiency, which is compared to a base case, and the results are refined through an iterative optimization loop to maximize injectivity and storage efficiency.

Our study highlights the critical necessity of automating the optimization of CO₂ injector placement, as suboptimal placement can have far-reaching consequences, including compromised CO₂ containment, reduced storage efficiency, and ultimately, jeopardized project viability. To mitigate these risks, the proposed workflow was rigorously tested using a dynamic reservoir simulator on a synthetic saline aquifer CO₂ storage complex model. The results unequivocally demonstrate the workflow's efficacy in optimizing the number and strategic placement of wells within the reservoir, predicting significant improvements in storage efficiency. Notably, when applied to the synthetic saline aquifer model, the automated workflow predicted a significant increase in storage efficiency compared to the reference case, highlighting its substantial potential to enhance CO₂ storage outcomes and contributing meaningfully to more effective and sustainable carbon sequestration strategies.

This innovative workflow has the potential to revolutionize the CCS industry by providing a data-driven, optimized approach to well placement and completion. By leveraging advanced simulation techniques, we can unlock the full potential of saline aquifers for CO₂ sequestration, supporting a more sustainable and environmentally conscious clean energy future.

Introduction

Global CO₂ emissions in 2024 increased by 0.9% to 36.3 Gt CO₂ (Deng et al. 2025), contributing to the growing concern of climate change. Greenhouse gases, such as CO₂, methane, and nitrous oxide, trap heat in the atmosphere, leading to global warming and associated environmental disruptions. Therefore, it is essential to reduce greenhouse gas emissions to mitigate the adverse effects of climate change. Carbon Capture and Storage (CCS) has emerged as a vital solution to achieve this goal, offering a promising technology to capture and store CO₂ emissions, thereby reducing the amount of greenhouse gases released into the atmosphere. CCS involves capturing CO₂ emissions from power plants and industrial processes, transporting them to a storage site, and injecting them into geological formations where they can be safely stored for thousands of years. Saline aquifers have emerged as a promising solution for large-scale CO₂ sequestration, offering significant storage potential due to their widespread availability and large storage capacity. However, the complex interplay of geological factors, including reservoir heterogeneity, spill point depth, and reservoir pressure, poses a significant challenge in optimizing well placement and completion strategies for CCS operations. The optimization of well placement is critical to the success of CCS projects, as suboptimal placement can have far-reaching consequences, including compromised CO₂ containment, reduced storage efficiency, and ultimately, jeopardized project viability. The practice of well placement using the traditional interpretation of static and dynamic data is time-consuming, subjective, and prone to errors, particularly in the context of increasingly complex CCS operations. As projects evolve and demand higher levels of storage efficiency while minimizing risks, there is a growing need for more advanced and automated workflows in well placement and completion. These workflows must integrate diverse data sources and leverage dynamic modeling to improve accuracy, consistency, and decision-making in real time.

In recent years, significant advances have been made in the development of simulation-based approaches for optimizing CCS operations. These approaches have shown great promise in improving storage efficiency, reducing risks, and enhancing overall CCS effectiveness. However, the integration of static and dynamic simulations to optimize well placement and completion strategies remains a significant challenge. This study seeks to overcome the challenges associated with optimizing CO₂ injector well placement in saline aquifers by developing an automated workflow that combines static model properties and dynamic simulations to determine optimal well locations and completion strategies for CCS operations in saline aquifers. The workflow is designed to identify optimal well locations and completion strategies by considering a range of critical variables, including:

- High-permeability zones and layers to facilitate efficient CO₂ injection
- Lateral and vertical continuity of the reservoir to ensure connected volume and minimize leakage risks
- Optimal well spacing to prevent interference and maximize storage capacity
- Placement of wells above the lowest closed contour (LCC) or spill point depth to ensure secure storage
- Pressure hotspots to avoid over-pressurization and ensure safe injection operations

By automating the optimization of CO₂ injector well placement, this workflow aims to enhance storage efficiency, minimize well injectivity issues, and improve overall CCS effectiveness.

Methodology

This workflow integrates reservoir and well variables in a dynamic and automated manner, enabling the optimal placement and completion of wells in areas with favorable reservoir properties, including high permeability, connectivity, and depleted pressure zones, which enhances the injectivity of CO₂ injectors and allows for the occupation of a larger pore volume, ultimately leading to improved storage efficiency. In order to identify these favorable areas for well locations, we utilize a CCS sweet spot tool that estimates the Reservoir Opportunity Index (R_{OI}) (Camargo 1999) a quantitative metric that evaluates the suitability of a reservoir for CO₂ storage, thereby enabling the precise identification of optimal well locations and completion strategies. The R_{OI} , which combines key reservoir variables such as oil saturation, hydrocarbon pore volume, and flow capacity to provide a comprehensive assessment of reservoir quality, as illustrated in Equation 1, allowing us to pinpoint the most promising areas for future well locations and optimize CO₂ storage operations.

$$R_{OI} = \sqrt[8]{I_0 I_c I_{kh}} \quad [1]$$

Where:

R_{OI} : Reservoir Opportunity Index

I_0 : Oil Saturation Index

I_c : Hydrocarbon Pore Volume Index

I_{kh} : Flow Capacity Index

The Reservoir Opportunity Index (R_{OI}) has evolved over time, with notable modifications by Shedid (2002) that incorporated additional parameters such as hydrocarbon pore volume, mobile oil saturation, reservoir pressure, and Rock Quality Index. To adapt the R_{OI} for Carbon Capture and Storage (CCS) applications, the original formulation was refined by replacing oil and hydrocarbon pore volume and saturations with water pore volume and saturation, as presented in Equation 2. The RQI is a quantitative metric that describes the porosity and permeability variations within different lithologies, identifying rocks with similar fluid flow characteristics, known as flow units, as illustrated in Equation 3.

$$SWPV = DX * DY * DZ * PORO * SW \quad [2]$$

Where:

SWPV: Water Pore Volume

DX, DY, DZ: Grid Simulation Dimensions

PORO: Porosity [unitless]

SW: Water Grid Saturation (at the relevant time step)

$$RQI = 0.0314 * \sqrt{\frac{PERM}{PORO}} \quad [3]$$

Where:

RQI: Reservoir Quality Index

PERM: Permeability [mD]

PORO: Porosity [unitless]

Three distinct Reservoir Opportunity Index (R_{OI}) equations can be employed, depending on the input parameters and desired output. The first equation, R_{OI}^1 , estimates reservoir quality based solely on the Rock Quality Index (RQI) and water saturation (SW), as illustrated in Equation 4. The second equation, R_{OI}^2 , refines the R_{OI}^1 estimate by incorporating the lowest closed contour (LCC) region, when provided, as shown in Equation 5. The third equation, R_{OI}^3 , builds upon R_{OI}^2 by introducing a desirable pressure threshold, as presented in Equation 6. And lastly, the fourth equation, R_{OI}^4 , builds upon R_{OI}^1 by introducing a desirable pressure threshold, as presented in Equation 7.

$$R_{OI}^1 = \sqrt{RQI * SWPV}, \text{ normalized between 0 and 1} \quad [4]$$

Where:

R_{OI}: Reservoir Opportunity Index

RQI: Reservoir Quality Index

SWPV: Water Pore Volume

$$R_{OI}^2 = ROI_1 \text{ ONLY in LCC Region selected Index (E.g. 0 or 1 or 2, etc)} \quad [5]$$

$$R_{OI}^3 = R_{OI}^2 \text{ ONLY in Pressure array selected Threshold (E.g. if Pressure < 3000 psi)} \quad [6]$$

$$R_{OI}^4 = R_{OI}^1 \text{ ONLY in Pressure array selected Threshold (E.g. if Pressure < 3000 psi)} \quad [7]$$

Following the identification of favorable areas, an automated workflow is employed to streamline the process of manually placing and completing CO₂ injectors in optimal locations, ensuring their injectivity remains uncompromised. This workflow is designed to accelerate the optimization of well placement and completion, and is outlined in [Figure 1](#), which illustrates the step-by-step process involved in achieving optimal storage efficiency. The workflow involves a series of integrated steps, including dynamic simulation, reservoir modeling, and optimization, which are iteratively refined to achieve the best possible results. By automating this process, the workflow enables the rapid identification of optimal well locations and completion strategies, allowing for the efficient and effective deployment of CO₂ injectors.

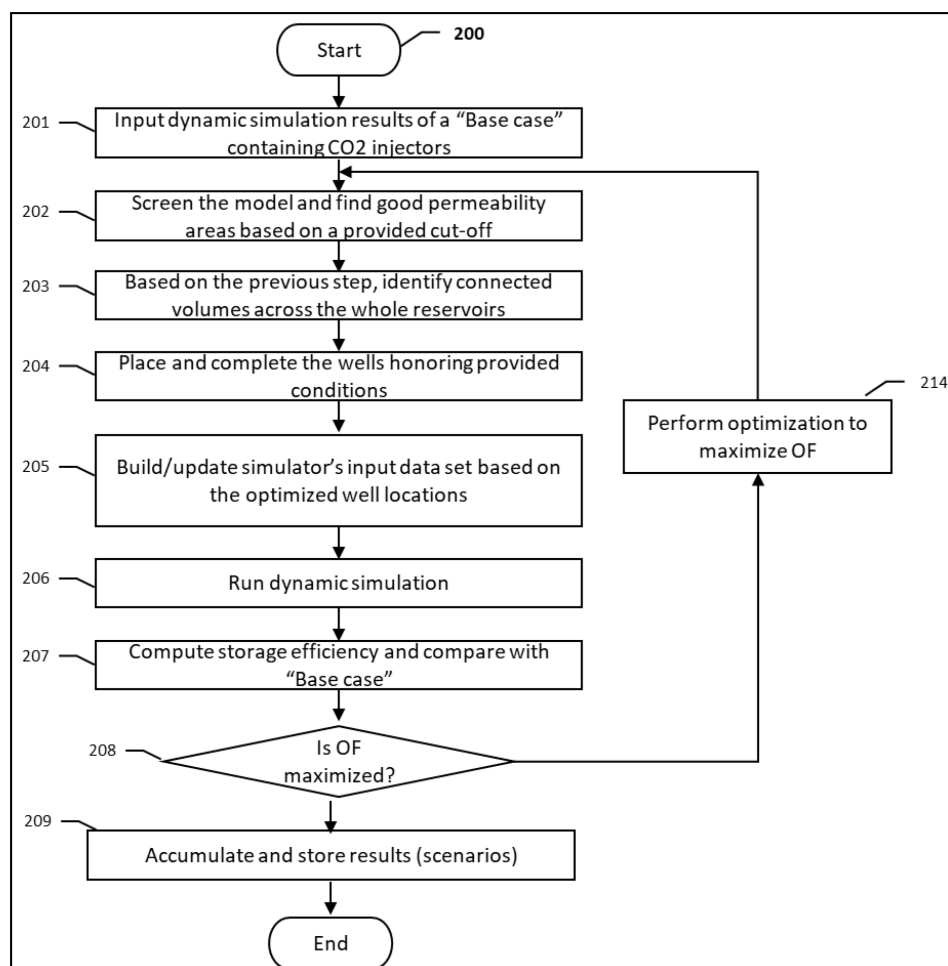


Figure 1—Schematic overview of the automated workflow for optimizing CO₂ storage well placement, which combines static and dynamic simulation to identify optimal well locations and completion strategies.

To test this work, 3D numerical model of an anticline structure, representing a finite aquifer, was built to perform dynamic simulation studies related to CO₂ underground storage using synthetic data and our in-house simulator GigaPOWERS. The aquifer structure, as shown in Figure 2, is 22.8 km in length and 15.5 km in width, and has a thickness of 258 m. The reservoir model consisted of a large grid with a total of approximately 9.6 million cells, discretized into 183 cells in the X-direction, 124 cells in the Y-direction, and 423 cells in the Z-direction, providing a highly detailed and comprehensive representation of the reservoir's geometry and properties.

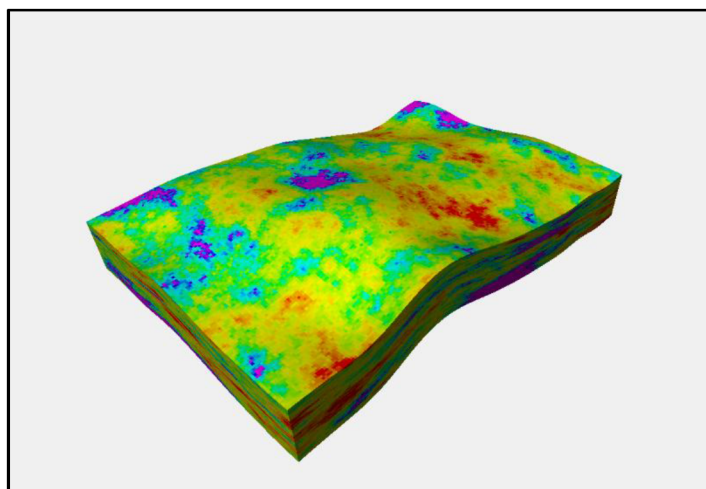


Figure 2—3D simulation model showing permeability distribution.

Model grid properties, like porosity and permeability, were generated by using random heterogeneity seeds. Normal distribution was used for porosity values of the population while permeability distribution was assumed to be correlated with porosity using log-normal distribution. Average values for static reservoir properties, can be found in Table 1.

Table 1—Schematic The key synthetic reservoir model parameters

Parameter	Value
Avg. Porosity	0.17
Avg. Permeability	25 mD
kv/kh ratio	0.10
Rock Compressibility	3.5×10^{-6} psi ⁻¹

A Black Oil model was used with two phases defined (GAS-WATER). The water phase was assumed to be a slightly compressible fluid with viscosity and density depending on the pressure change. The reservoir was assumed to be fully saturated with water initially and in an isothermal condition. CO₂ dissolution in the water phase was not considered in the study.

Results & Discussion

Base Case Results

The base case simulation was designed to establish a reference scenario for the underground CO₂ storage process. In this scenario, 12 vertical CO₂ injector wells were randomly distributed across the structure to maximize the utilization of the available pore volume and achieve a target injection rate. The well placement

is illustrated in Figure 3. Each well was completed across layers 38 to 417 of the reservoir, ensuring a comprehensive coverage of the storage formation.

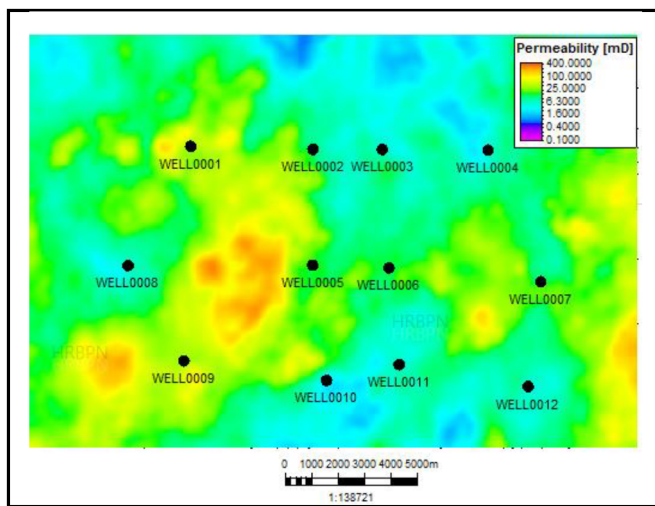


Figure 3—Average permeability map showing CO₂ injectors randomly distributed.

The operational scheme for the base case simulation consisted of a generic CO₂ injection strategy, which involved injecting 12 million metric tons per annum (MMTPA) of CO₂ over a period of 20 years, followed by a 55-year monitoring phase to track the migration of the CO₂ plume. The field had an initial reservoir pressure of around 3000 pounds per square inch (psi). The injection wells were constrained to operate within specific limits, with a maximum injection rate of 2 MMTPA (approximately 104 million standard cubic feet per day, MMSCFD) and a maximum bottom hole pressure of 4250 (psi).

The results of the base case simulation are presented in Figure 4, which shows the CO₂ injection rate plateau for the reference scenario with random well placement. The simulation indicates that the target injection rate is met for less than 2 years before the injection rate begins to decline. This decline is primarily attributed to the increasing reservoir pressure, which approaches the maximum allowable bottom hole pressure of 4250 psi. As the pressure limit is reached, as shown in Figure 5, the simulated injection rate per well reduces, resulting in a decrease in the overall CO₂ injection rate. The base case simulation highlights the importance of managing reservoir pressure to maintain a stable and efficient CO₂ injection process. The results of this simulation will serve as a reference point for evaluating the effectiveness of alternative well placement strategies and operational schemes.

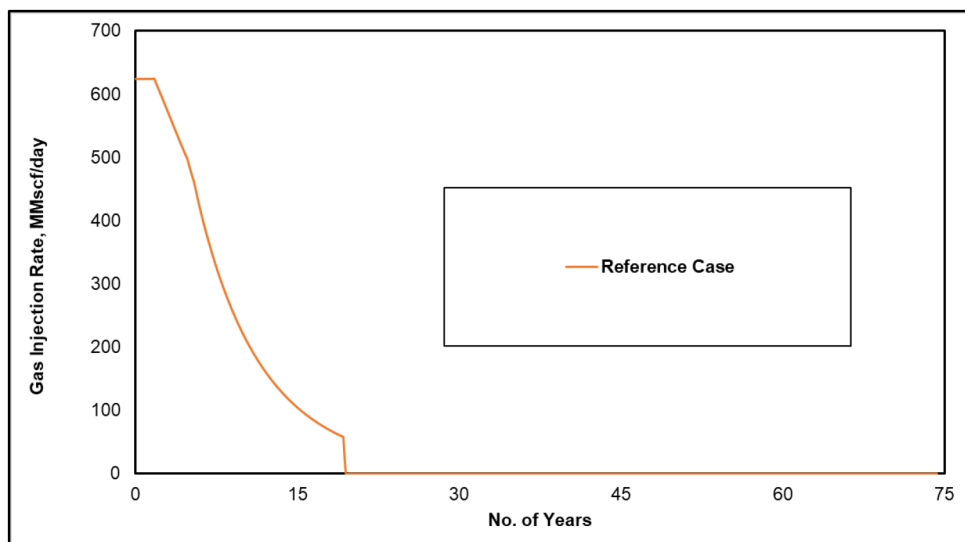


Figure 4—The CO₂ injection profile of the reference case.

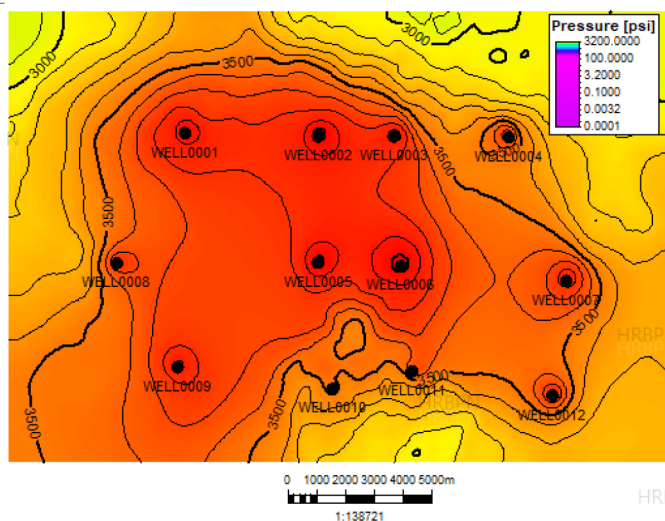


Figure 5—Average reservoir pressure map at year 2031 illustrating CO₂ randomly distributed.

Optimized CO₂ Injectors Well Placement: Results and Comparison

The application of the automated workflow and tools enabled the identification of optimal locations for the CO₂ injectors, leveraging the "sweet spot" concept to maximize injection efficiency. The new injector locations, illustrated in Figure 6, are strategically positioned on an average reservoir opportunity index map, which highlights the most favorable areas for CO₂ injection. The optimized well placement considers the interconnected volumes, and reservoir pressure ensuring that the injectors are situated in areas with the highest potential for efficient CO₂ storage.

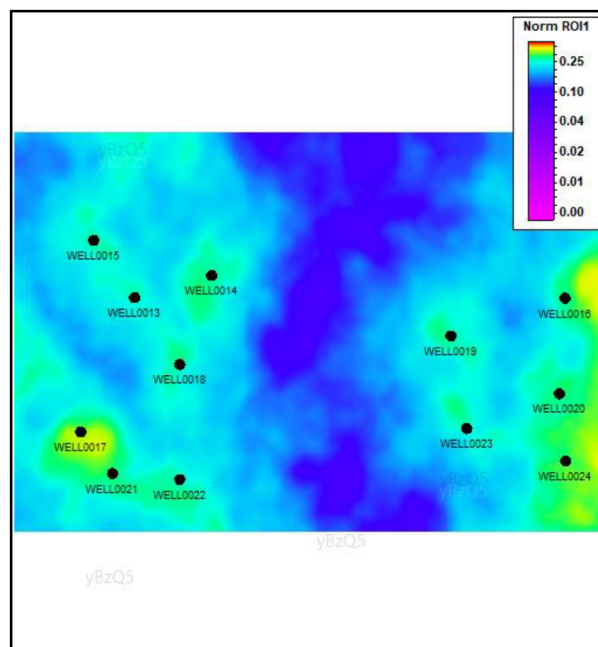


Figure 6—Average reservoir opportunity index (R'_{oi}) map illustrating CO₂ injectors targeting sweet spots.

Figure 7 provides further insight into the optimized well placement, showing the locations of the injectors on an average reservoir pressure map after 6 years of injection. The results demonstrate that the new well placement strategy effectively targets areas with favorable reservoir pressure, minimizing the risk of pressure buildup and maximizing the potential for long-term CO₂ storage.

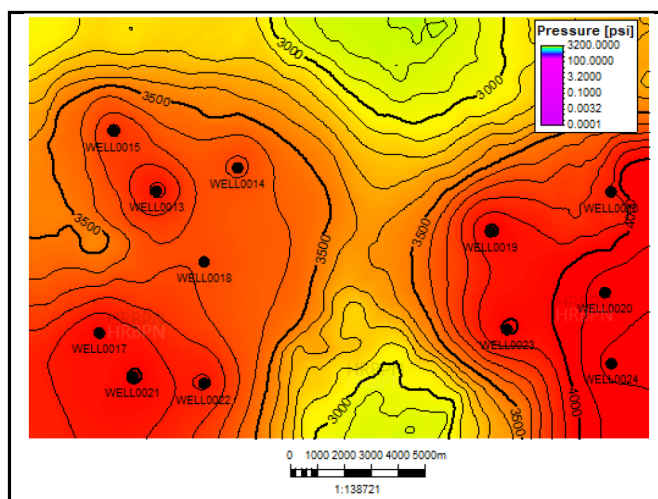


Figure 7—Average reservoir pressure map at year 2031 illustrating CO₂ injectors targeting sweet spots.

A comparison of the CO₂ injection plateaus between the reference case and the optimized case is presented in Figure 8. The results show a significant improvement in the optimized case, with the CO₂ injection plateau increasing by more than 5 years. This substantial increase in plateau duration is a direct result of the optimized well placement strategy, which enables more efficient use of the reservoir's storage capacity and reduces the pressure buildup associated with CO₂ injection. The optimized well placement strategy demonstrates the potential for improved CO₂ storage efficiency and highlights the value of using advanced workflows and tools to inform well placement decisions.

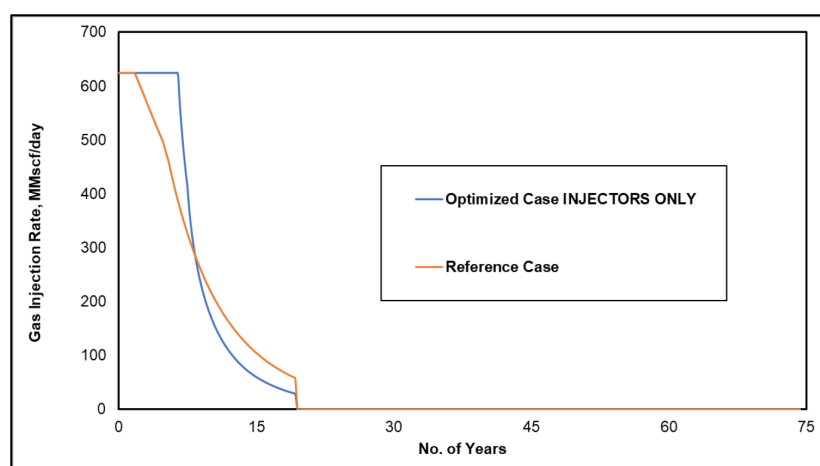


Figure 8—The CO₂ injection profile of the reference case and the optimized case with injectors only.

Enhanced Optimization with Relief Wells: Meeting the 20-Year Plateau Target

To further optimize the CO₂ injection process and meet the target plateau for the entire 20-year injection period, a relief system was integrated into the optimized case. This enhancement aimed to maximize pore volume utilization and achieve the target injection rate per well while adhering to the reservoir's fracture limit. Building on the concept presented by AlZayer (2023) for optimizing reservoir pressure management, we leveraged this innovative approach to design a strategic scenario that involved the targeted placement of 12 water relief wells. This carefully planned configuration was designed to effectively alleviate pressure buildup and facilitate increased CO₂ injection, thereby maximizing the ultimate CO₂ storage capacity.

The relief wells were carefully positioned using the same advanced tools and workflow employed for the optimized CO₂ injector well placement. This ensured that the relief wells were located in optimal positions to effectively manage pressure buildup and prevent CO₂ production. The relief wells were constrained to produce a maximum to 20k barrels per day (bpd). Figure 9 and 10 illustrates the placement of the relief wells in relation to the CO₂ injectors, demonstrating a well-designed system for pressure management.

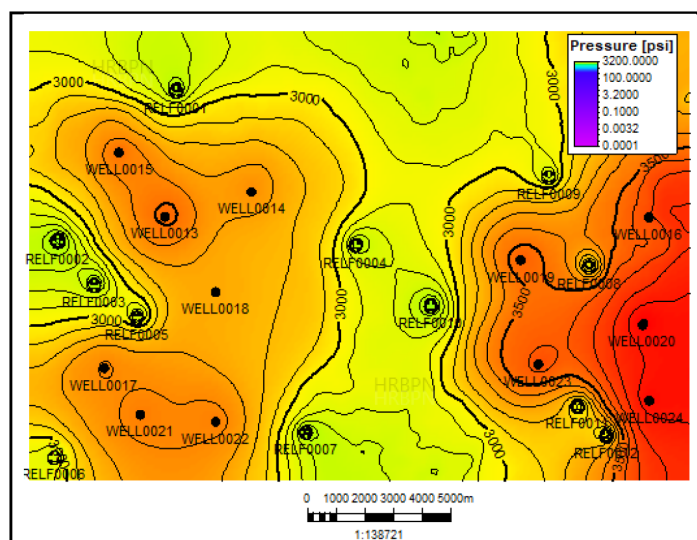


Figure 9—Average reservoir pressure map at year 2033 showing CO₂ injectors and relief wells targeting sweet spots.

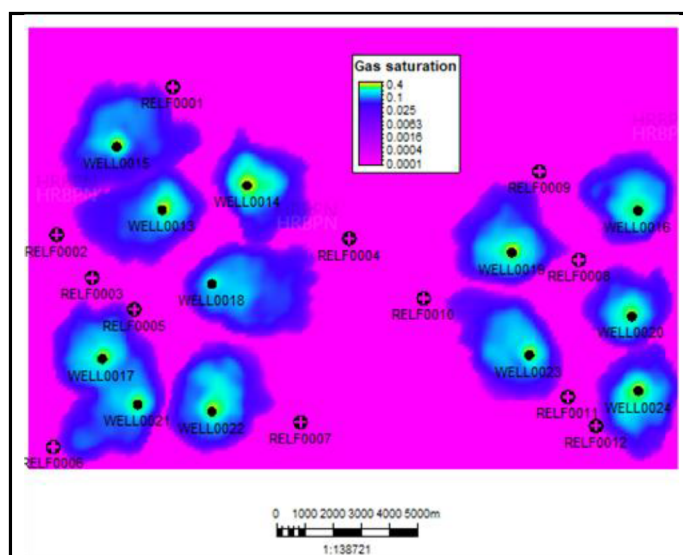


Figure 10—Average reservoir saturation map at year 2045 showing CO₂ injectors and relief wells targeting sweet spots.

The results of this enhanced optimization are presented in Figure 11, which shows that the CO₂ injection plateau has been successfully maintained for the full 20-year injection period, followed by a 55-year monitoring phase. Importantly, Figures 10 and 12 also confirm that the relief wells have operated as intended, with no CO₂ production observed. This is a critical outcome, as it indicates that the injected CO₂ has been effectively stored in the reservoir, with no leakage or migration to the relief wells.

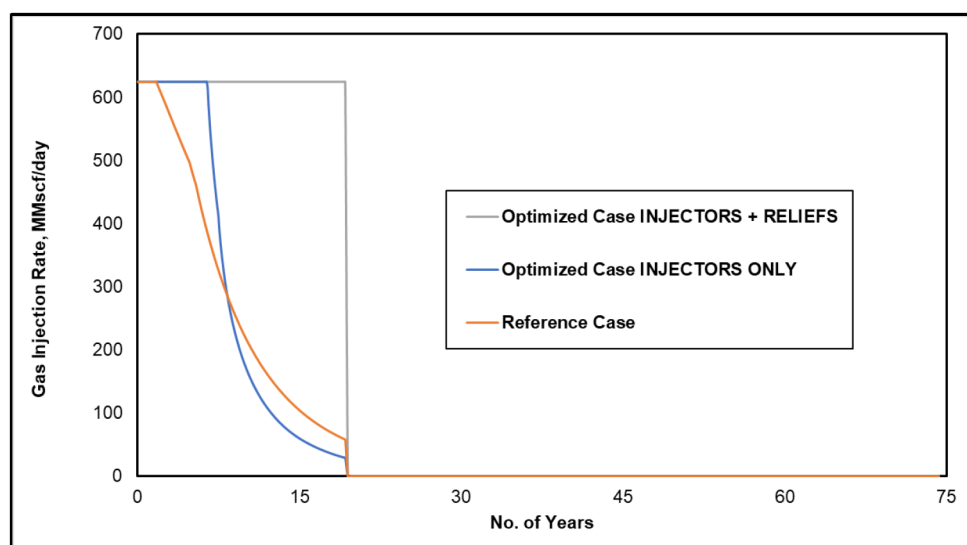


Figure 11—The CO₂ injection profile of the reference case, optimized case with injectors only, and optimized case with injectors and relief.

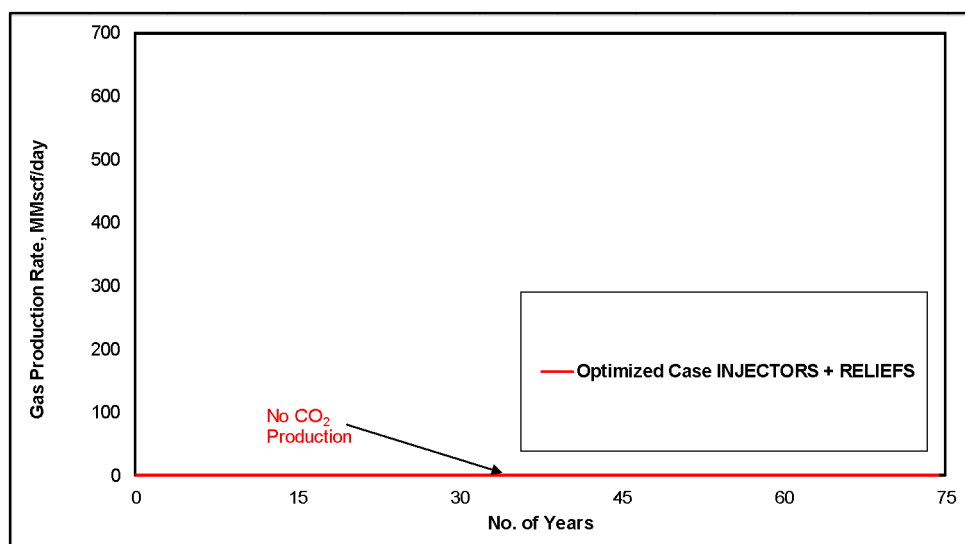


Figure 12—The CO₂ production profile of the optimized case with injectors and relief.

Validation of Workflow Reliability: Storage Efficiency Comparison

To further validate the reliability and effectiveness of the optimized workflow, a comprehensive comparison of the storage efficiency for each case was conducted. Storage efficiency is a critical metric that quantifies the amount of CO₂ stored in the reservoir relative to the available pore volume. As illustrated in Figure 13, the results show a significant increase in storage efficiency for both optimized cases compared to the reference case. This improvement is a direct consequence of the optimized well placement and pressure management strategies employed in the workflow.

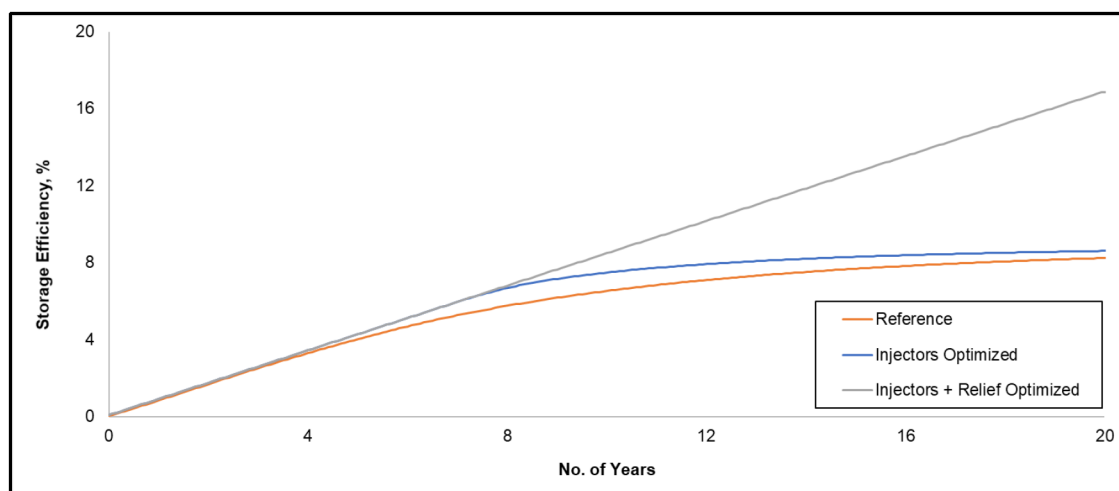


Figure 13—The storage efficiency profile of the reference case, optimized case with injectors only, and optimized case with injectors and relief.

Notably, the combination of the optimized workflow with a pressure relief system yields a remarkable enhancement in storage efficiency, increasing from around 8% to almost 17%. This substantial improvement demonstrates the potential of the workflow to substantially enhance the effectiveness of CO₂ storage operations. By maximizing the utilization of the available pore volume and minimizing pressure buildup, the optimized workflow enables more efficient storage of CO₂, ultimately contributing to a more successful and sustainable carbon sequestration project. The comparison of storage efficiencies across the different

cases provides strong evidence for the reliability and validity of the workflow, highlighting its potential to inform and optimize CO₂ storage operations.

Conclusion

In conclusion, this study demonstrates the critical importance of optimizing CO₂ injector placement and completion in saline aquifers to maximize storage efficiency and minimize well injectivity issues. The developed automation workflow, which integrates static model properties and dynamic simulations, has been shown to be highly effective in identifying optimal well locations and completion strategies. By leveraging this workflow, we have achieved a significant increase in storage efficiency, with results showing a significant increase in storage efficiency compared to the reference case. This substantial enhancement in storage efficiency is a direct result of the optimized well placement and completion strategies, which consider key geological factors such as reservoir heterogeneity, and pressure buildup. Notably, the well placement tool developed in this study is versatile and can be applied to both CO₂ injectors and relief wells, allowing for the optimization of entire CO₂ storage systems. The ability to strategically place relief wells in conjunction with injectors enables the effective management of reservoir pressure, further enhancing storage efficiency and reducing the risk of well injectivity issues. The findings of this study underscore the importance of careful planning and optimization in CO₂ storage operations, and demonstrate the potential for advanced simulation techniques to unlock the full potential of saline aquifers for CO₂ sequestration. Ultimately, this research contributes to the advancement of carbon sequestration technologies and supports the transition to a low-carbon economy, aligning with the urgent need to reduce greenhouse gas emissions and mitigate climate change.

Reference

- Deng, Z., Zhu, B., Davis, S.J., et al 2025. Global carbon emissions and decarbonization in 2024. *Nat Rev Earth Environ* **6**, 231–233. <https://doi.org/10.1038/s43017-025-00658-x>
- Camargo A., 1999. "Prioritizing opportunities for new well locations and well workovers." *Geoquest Forum*.
- Amaefule, J., Altunbay, M., Tiab, D., et al 1993."Enhanced Reservoir Description: Using Core and Log Data to Identify Hydraulic (Flow) Units and Predict Permeability in Uncored Intervals/Wells." Paper presented at the SPE Annual Technical Conference and Exhibition. doi: <https://doi.org/10.2118/26436-MS>
- Shedid, Shedid A., and Reyadh A. Almehaideb. 2002. "A New Approach of Reservoir Description of Carbonate Reservoirs." Paper presented at the SPE International Petroleum Conference and Exhibition. doi: <https://doi.org/10.2118/74344-MS>
- Al Zayer, H., Saeed, M. 2023. "Optimizing Reservoir Pressure Management with Relief Wells During CO₂ Storage in Finite Aquifers." Paper presented at the Middle East Oil, Gas and Geosciences Show. doi: <https://doi.org/10.2118/213558-MS>